

Platform Competition in the Autobidding World

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Abstract

We study the problem of auction design for advertising platforms that face strategic advertisers who are bidding across platforms. Each advertiser's goal is to maximize their total value or conversions while satisfying some constraint(s) across all the platforms they participates in. In this paper, we focus on advertisers with return-over-investment (henceforth, ROI) constraints, i.e. each advertiser is trying to maximize value while making sure that their ROI across all platforms is no less than some target value. An advertiser interacts with the platforms through autobidders – for each platform, the advertiser strategically chooses a target ROI to report to the platform's autobidder, which in turn uses a uniform bid multiplier to bid on the advertiser's behalf on the queries owned by the given platform.

Our main result is an auction comparator theorem that can be used to choose between different auction mechanisms by a revenue-maximizing platform operating in a competitive environment. Analogous to the Linkage Principle for interdependent values, our comparator demonstrates that, for a broad class of advertiser valuations, second-price auctions may sometimes generate higher revenue than first-price auctions. This contrasts sharply with the single-platform setting, where first-price auctions are optimal for both revenue and welfare [14]. Our findings highlight the critical importance of considering inter-platform competition when designing optimal auctions.

CCS Concepts

• Applied computing → Online auctions.

Keywords

Online auctions, autobidders, multiple platforms, platform design, uniform bidding

ACM Reference Format:

Gagan Aggarwal, Andres Perlroth, Ariel Schwartzman, and Mingfei Zhao. 2026. Platform Competition in the Autobidding World. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United

Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792225>

1 Introduction

Online advertisers often optimize their advertising campaigns across different advertising platforms like Amazon, Bing, Google, Meta and TikTok to reach their audience effectively. Each platform, driven by its own goals, strategically designs its auction mechanism to sell ad space. While existing research provides valuable insights on auction design within individual platforms, understanding how competition between platforms influences auction design remains largely unexplored.

This question becomes critical when advertisers aim to maximize their value (e.g. brand awareness, conversions, etc.) subject to a minimum ROI constraint they need to meet across platforms. In this case, advertisers' investment in a platform is sensitive to the value over cost the platform provides: a platform with a more price-intense auction format is less attractive from an advertisers' standpoint. Therefore, the auction design of one platform creates externalities for the other platforms through the constraints of the advertisers.

In this paper, we study the question of auction design for competing platforms that face strategic advertisers bidding across the platforms. Advertisers interact with the platforms using autobidders that bid on their behalf on each query. Autobidders bid using a *uniform* bidding algorithm whereby the bid on a query is proportional to the advertiser's value for the query.¹ Our main result is an auction comparator theorem that, akin to the Linkage Principle in the context of auctions with interdependent values [17], can be used to choose between two auctions by a platform in a competitive environment. Our comparator shows that with sufficient competition across platforms, a platform often benefits from using a Second Price Auction (SPA) mechanism instead of using a First Price Auction (FPA). This is in stark contrast to the case of a single platform where FPA is the revenue-optimal auction when advertisers are value-maximizing agents [14].² Our paper, therefore, shows



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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792225>

¹Uniform bidding, also known in the literature as bid-pacing, are well-studied algorithms in the literature as their simplicity and good performance make them appealing from a practical standpoint [8, 9, 14]

²Observe that our result does not imply that a platform using FPA is acting as monopolist in the market – there are other reasons why a platform would use FPA that our theory does not cover. For example, in Display Ads, first-price auctions are widely used because they are credible [6] and increase spending confidence of advertisers [10, 15] as they are charged by their own bid.

that competition across platforms is a key factor to consider when choosing an auction format.

Our multi-platform auction comparator captures the main trade-off between more aggressively priced auctions (like FPA) and less aggressively priced ones (like SPA). On the one hand, because advertisers are value-maximizing agents with ROI constraints and bid uniformly, their shading is weaker than with profit-maximizing agents and therefore FPA is more effective in extracting revenue. On the other hand, when advertisers optimize across multiple platforms, their per-platform bids are sensitive to pricing differences between the different platforms, leading to higher bid shading on more aggressively-priced auctions formats like FPA. More concretely, with multiple platforms, advertisers bid so as to equalize the marginal ROI across platforms. One can show that when the platform chooses SPA, the marginal ROI at a given bid level is higher than when the platform chooses FPA. Therefore they will shade their bids on FPA and increase their bids on SPA. Our auction comparator theorem shows that when the difference of the marginals is large (which is endogenously determined) and the competition is intense, a platform prefers SPA over FPA.

Our model consists of multiple revenue-maximizing platforms each owning multiple queries. Each platform independently chooses a per-query auction mechanism (e.g., SPA, FPA) and sells each of their queries according to the chosen auction.³ Each advertiser is a value-maximizing agent with a target ROI constraint that upper-bounds the average cost per unit value (e.g., conversions) they can pay across all the platforms they participate in. For each advertiser-platform pair (i, j) , advertiser i has value v_{ij} for queries on platform j , which is drawn independently from some public distribution D_{ij} . For simplicity, we will abstract away the role of the autobidders from the model. In practice, an advertiser might report a platform-specific target to an autobidder, who would then computes a uniform bid multiplier to meet this target. We bypass this step by allowing the advertiser to directly (and strategically) choose their own bid multiplier for each platform. This allows us to simplify the strategy space of the advertisers to the set of bid multipliers they could submit to each platform, denoted as $\{\mu_{ij}\}_j$.

The platforms and advertisers play a sequential game as follows:

- (1) First, all platforms independently choose a per-query auction to sell their queries.
- (2) Second, the advertisers strategically choose their uniform bid multipliers μ_{ij} for every platform j they participate in.
- (3) For each query on platform j , each advertiser i draws her value $v_{ij} \sim D_{ij}$ and submits a bid of $\mu_{ij} \cdot v_{ij}$ for the query.
- (4) Finally, queries are allocated and payment accrues according to the chosen auction format.

We make the following assumption which is common in the autobidding literature [2, 3, 5, 7, 14]: advertisers' target ROIs and their value distributions are common knowledge to all agents in the game. As discussed in previous literature, the high frequency of ad-auctions allow autobidders to have sharp estimates of competitors' bids, which conceptually mimics the theoretical framework of complete information games. For the solution concept, we restrict

³In the paper, we make some mild restrictions on the auction class: the highest bidder always win the auction; the winner never pays more than the bid; losing bidders do not pay; payment scales linearly with bids (i.e. the payment rule as function of bids is homogeneous of degree 1).

to subgame perfect equilibrium so as to fully capture the dynamics that have to be taken into account for the platform's auction decision: how advertisers respond to auction changes by changing the uniform bid multipliers submitted to platforms.

In Section 3, we first solve the advertisers' subgame and characterize optimal bidding given platforms' auction choices. Theorem 3.2 shows that advertisers choose bid multipliers such that they obtain the same *marginal cost* across all platforms. Here marginal cost is the advertiser's additional cost for getting a unit of additional value. We then show how the marginal cost an advertiser obtains on a platform is directly linked to the aggressiveness of the pricing rule of the auction. When the auction is SPA, the marginal cost on that platform is exactly the bid multiplier itself that the advertiser chooses. On the other hand, if the platform uses FPA, then the marginal cost is strictly larger, and depends on the environment that the advertiser faces (i.e. how other advertisers bid) on the platform.

In Section 4, we develop the first version of our auction comparator result – it applies to settings where all advertisers are symmetric, i.e. $D_{ij} = D_{i'j}$ for all platforms j and pairs of advertisers i, i' . We notice that at any symmetric advertiser equilibrium, the allocation is always efficient regardless of the platforms' auction choices, i.e. the advertiser with the highest value always win the auction.

THEOREM 1.1 (AUCTION COMPARATOR THEOREM). *Let $\mathcal{E}(A)$ be the ratio between the marginal cost and the bid multiplier at the efficient allocation. Let $R(A)$ be the revenue of a platform when using auction A if all advertisers bid their value. When advertisers are symmetric, a platform prefers auction A over auction A' regardless of the other platforms' choices if and only if*

$$\frac{R(A)}{\mathcal{E}(A)} > \frac{R(A')}{\mathcal{E}(A')}.$$

Because advertisers equalize marginal cost across platforms, the term $\mathcal{E}(A)$ can be thought of as a form of "elasticity" that captures the magnitude of bid shading (relative to bidding their value), due to the aggressiveness of the pricing of auction A relative to other platforms. The larger the elasticity $\mathcal{E}(A)$ is, the more costly the auction is, resulting in a smaller bid multiplier on the platform. Since the revenue in the auction scales linearly with bids (see Theorem 2.1), the net revenue impact of using auction A is the level of bid shading (i.e. the bid multiplier), captured by the elasticity terms, times the revenue when advertisers bid their value.

It's worth pointing out that Theorem 1.1 requires only the advertisers to be symmetric and allows platforms to be asymmetric. This captures the setting where the platforms have identical inventories but asymmetric market share, i.e. $D_{ij} = \alpha_j \cdot D$ for some distribution D , $\forall i, j$, where α_j is the market share of platform j .

We then apply our auction comparator to fully solve the equilibrium for the case where the advertisers are symmetric and, in particular, to characterize cases where FPA (the optimal auction in the single platform model) is no longer the optimal auction.

THEOREM 1.2. *Assume advertisers are symmetric. Let $\mathcal{A}$ be the class of feasible auctions that the platforms can use. Then, in any equilibrium, platform j selects A_j^* such that*

$$A_j^* = \arg \max_{A \in \mathcal{A}} \frac{R_j(A)}{\mathcal{E}_j(A)}.$$

In particular if $\mathcal{A} = \{SPA, FPA\}$, then SPA strictly dominates FPA if and only if $\mathcal{E}_j(FPA) \cdot C_j > 1$, where $C_j = R_j(SPA)/R_j(FPA)$ measures the intensity of the competition across advertisers in the platform.

In Section 5, we focus on settings where advertisers are asymmetric. In these settings, the equilibrium allocation may not always be efficient. This makes solving the general setting completely very challenging as the elasticity $\mathcal{E}(A)$ depends on the equilibrium allocation, and thus depends on the other platforms' actions and on the autobidders' bidding equilibrium in each of the platforms. We, nonetheless, provide sufficient conditions for SPA to dominate FPA as well as sufficient conditions for FPA to dominate SPA. Intuitively, since FPA with uniform bidding allocates queries more efficiently than SPA, the conditions of Theorem 1.2 (appropriately modified for the general setting) should be sufficient for FPA to be selected at equilibrium. Indeed, in Theorem 5.1, we generalize the conditions of Theorem 1.2 for settings with asymmetric advertisers, and show that when these conditions are met, FPA is selected in equilibrium. On the other hand, for SPA to be selected at equilibrium, the bidding response induced by providing better marginal cost to advertisers needs to be strong enough to compensate for the loss in allocation efficiency induced by using SPA. Thus, the conditions of Theorem 1.2 (appropriately modified) are now necessary but not sufficient. Theorem 5.2 provides a stronger condition for SPA to be chosen at equilibrium: $\mathcal{E}_j(FPA) \cdot C_j > f(\text{Inefficiency due to SPA}) > 1$ for a specific function f detailed in Section 5. We remark that both conditions are considered as a generalization of Theorem 1.2 in the asymmetric-advertiser setting in the sense that both conditions reduce to the exact condition in Theorem 1.2 when advertisers are symmetric.

We instantiate our results for a parametrized class of instances for two advertisers where one advertiser has uniform distribution and the other has a point mass distribution. We find the subgame perfect equilibrium when platforms choose between FPA and SPA. We also provide some complementary numerical analysis to understand how the market share of platforms might shape their auction decisions. Our simulations show that as the market becomes more concentrated, the dominant platform is more likely to use FPA while as the competition becomes more intense, SPA becomes the preferred strategy (see Section 6.1).

Finally, we present two results that showcase the importance of the advertisers' elasticity in allowing for equilibria other than the one where all platforms use FPA. In the first result, we force all advertisers to use the same bid multiplier across all platforms. In the second result, there is a single *strategic* advertiser bidding across platforms and competitors are *static* advertisers that always bid their value. For each of the results, we disable the advertisers' reaction to an auction change to some degree, and find that the platforms always end up in the all-FPA equilibrium (see Section 7).

1.1 Related Work

With the growth of autobidding in online ad auctions, there has been substantial research interest in problems related to autobidders. Aggarwal et al. [2] provides a thorough survey on the topic. In what follows, we discuss specific papers that relate to our work.

Multi-Channel Mechanism design. Aggarwal et al. [5] focuses on a single platform owning multiple channels, each selling their queries using SPA with a reserve price. The authors quantify the cost of having each channel optimizing its own reserve price compared to a global platform policy. In our paper, we study competition across platforms and consider the problem of platforms choosing between different auction mechanisms, not just optimizing their reserve prices. Motivated by the Display Ad market, Paes Leme et al. [18] studies a model where multiple platforms compete for profit-maximizing bidders that are constrained to use the same bid on all platforms (in our language, they bid using a uniform bid). Their main result shows that FPA is the optimal auction for the platforms, similar to Theorem 7.2 where we show the same conclusion for the value-maximizing case.

Bidding Algorithms for Autobidders, Uniform Bidding.

Uniform bidding (aka constant pacing) has been extensively studied in the context of budget constraints [1, 11, 12, 20] and, more generally, for autobidding on a single platform. Aggarwal et al. [3] initiated the study of the autobidding problem and, among other things, showed that uniform bidding is optimal if the platform uses a truthful auction format. Liaw et al. [16] show that for non-truthful auctions uniform bidding is suboptimal and when autobidders use (optimal) non-uniform bidding, the revenue outcomes is no better than using a truthful auction. In this work, while we restrict advertisers to use uniform bidding within a platform, advertisers are allowed to do non-uniform bidding *across* platforms.

Bidding across Multiple Platforms. Susan et al. [19] studies bidding strategies for utility-maximizing advertisers across multiple platforms under budget constraints. Deng et al. [13] focuses on bidding for value-maximizing advertisers under both budget and ROI constraints and shows that optimizing over per-platform ROIs can be arbitrarily bad when the advertiser has both ROI and budget constraints. In contrast, with only ROI constraints, we show that marginal equalization using per-platform ROIs is the optimal bidding strategy. Finally, Aggarwal et al. [4] study the query complexity of finding the optimal bidding strategy in the multi-platform setting with and without prior information. Their result builds on the fact that an advertiser's optimal bidding strategy involves equalizing the marginal ROI across platforms, as stated in Theorem 3.2 of our paper.

2 Model

Our environment consists of two groups of strategic agents, platforms and advertisers.

Platforms: There are m platforms in the market and we denote $J = [m]$ the set of all platforms. Platform $j \in J$ owns a series of single-shot queries.

Each platform wants to maximize their revenue and chooses a per-query auction $A = (\mathbf{x}, \mathbf{p})$ to sell the queries following the given auction: given bids $\mathbf{b} = (b_1, \dots, b_n)$, bidder i wins if and only if $x_i(\mathbf{b}) = 1$ and pays a price $p_i(\mathbf{b})$, where x_i are non decreasing functions on b_i . Let $\mathcal{A}$ be the set of auction mechanisms that platforms can choose from. Considering the common auctions used in advertising, we assume that any auction in $\mathcal{A}$ satisfies the following natural properties.

DEFINITION 2.1. Any auction $A = (\mathbf{x}, \mathbf{p}) \in \mathcal{A}$ satisfies the following for any bids $\mathbf{b} = (b_1, \dots, b_n)$:

- (1) *Highest bid wins:* if $b_i > b_{i'}$ for all $i' \neq i$, then $\mathbf{x}_i(\mathbf{b}) = 1$. When there are multiple highest bids, we break ties lexicographically.⁴
- (2) *No bidder pays more than her bid:* $p_i(\mathbf{b}) \leq b_i$.
- (3) *Only the winning bidder pays:* if $\mathbf{x}_i(\mathbf{b}) = 0$ then $p_i(\mathbf{b}) = 0$.
- (4) *Payment $p_i(\mathbf{b})$ is weakly monotone on b_i .*
- (5) *Payments are homogeneous in bid:* for every $\lambda > 0$, $p_i(\lambda \cdot \mathbf{b}) = \lambda \cdot p_i(\mathbf{b})$. Note that by Property (1), $\mathbf{x}_i(\mathbf{b}) = \mathbf{x}_i(\lambda \cdot \mathbf{b})$.

Observe that canonical auctions like a second-price auction (SPA), first-price auction (FPA) or a linear combination of both auctions satisfy Definition 2.1.

Advertisers: There are n advertisers in the system and we denote $I = [n]$ the set of advertisers. Each advertiser $i \in I$ has value v_{ij} for winning a query in platform j . We assume that v_{ij} is drawn independently from some known distribution D_{ij} . We assume that D_{ij} has a positive density almost everywhere in its support (so that the CDF function is continuous) and $E_{v_{ij} \sim D_{ij}}[v_{ij}] < \infty$. For ease of notations, we denote v_j (or D_j) all advertisers' value profile (or its distribution) for a given platform j . We also denote v_{-i} (or D_{-i}) and $v_{-i,j}$ (or $D_{-i,j}$) the value profile (or distribution) for all advertisers' except i , for all platforms and a given platform j respectively.

Each advertiser i can choose a vector of bid multipliers $\{\mu_{ij}\}_j$. The bid submitted to each platform is proportional to the advertiser's value, i.e. a *uniform bid* within each platform. Formally $b_{ij} = \mu_{ij} \cdot v_{ij}$. We use $\mu_j, \mu_{-i}, \mu_{-i,j}$ for the bid multipliers.

The advertiser's objective is to maximize its total expected value across the platforms with a Return-on-Investment (ROI) constraint with target T_i over all platforms. Given the (payment rule of) auction of the platforms and the bid multipliers for other advertisers μ_{-i} , advertiser i aims to solve the following problem

$$\begin{aligned} \max_{\{\mu_{ij}\}_j} & \sum_{j \in J} E_{v_j \sim D_j} [v_{ij} \cdot \mathbb{1}[\mu_{ij} v_{ij} \geq \mu_{kj} v_{kj}, \forall k \neq i]] \\ \text{s.t. } & T_i \cdot \sum_{j \in J} E_{v_j \sim D_j} [p_{ij}(\mu_j \cdot v_j)] \leq \\ & \sum_{j \in J} E_{v_j \sim D_j} [v_{ij} \cdot \mathbb{1}[\mu_{ij} v_{ij} \geq \mu_{kj} v_{kj}, \forall k \neq i]] \end{aligned} \quad (1)$$

Observe that we can re-scale the values by defining $v'_{ij} = v_{ij}/T_i$. Thus we assume without loss of generality that $T_i = 1$ for all advertisers throughout the paper.

Timing, Information and Equilibria

The problem we are interested in studying is modeled by the following sequential game where platforms move first and advertisers respond.

- (1) Each platform j independently and simultaneously announces its respective per-query auction format.
- (2) Each advertiser i independently and simultaneously submits multipliers μ_{ij} for each platform j they bid on.

- (3) When a query q comes, advertiser realizes value v_{ij} drawn from D_{ij} independently, and submits bid $\mu_{ij} \cdot v_{ij}$.
- (4) Allocations and payments are calculated according to the auction rules.

We assume that the advertisers' targets T_{ij} and value distributions D_{ij} are common knowledge to all players, so that the game is of complete information.⁵ Our solution concept is subgame perfect Nash equilibrium (SPNE).

We impose the following equilibrium refinements on the advertisers' bidding subgame to remove unnatural equilibria and asymmetric equilibria in symmetric environments.

DEFINITION 2.2. Let $(\mu_{ij})_{i \in I, j \in J}$ be the bid multipliers in SPNE given the platforms' auction choices. We impose the following refinements:

- (R1) For every advertiser i , the multipliers $\{\mu_{ij}\}_j$ at equilibrium are not weakly dominated by another set of multipliers $\{\mu'_{ij}\}_j$.
- (R2) If two advertisers i, i' have the same distributions on every platform, i.e., $D_{ij} = D_{i'j}, \forall j \in J$, then in equilibrium, both advertisers use the same set of bid multipliers.
- (R3) If each advertiser i has the same distribution on two different platforms $j, j' \in J$, i.e., $D_{ij} = D_{ij'}, \forall i$, and both platforms use the same auction, then in equilibrium each advertiser uses the same bid multiplier on both platforms, i.e., $\mu_{ij} = \mu_{ij'}, \forall i$.

Refinement (R1) is standard in the literature (e.g., [3, 5]) and removes unnatural equilibria: for example, if there is a single platform running SPA, then the strategy where advertiser 1 uses an sufficiently large bid multiplier and the advertiser 2 bids 0 is an (unnatural) equilibrium: advertiser 1 always win the auction with 0 payment so her ROI constraint is satisfied and she can not get more value; advertiser 2 can never win since otherwise she has to suffer payment equals to advertiser 1's high value. However this equilibrium does not pass this refinement as for advertiser 2, using a bid multiplier of 0 is dominated by bid multiplier 1. Refinements (R2)-(R3) restrict to symmetric bidding equilibria whenever advertisers are symmetric or platforms are fully symmetric in terms of the advertisers' value distributions and auction choice.

In general there might be multiple advertiser equilibria when platforms decide their auction choices, which may cause the platforms to have different payoffs. Throughout the paper, when studying the game among platforms, we focus on the interior advertiser equilibrium where no advertiser wins the entire inventory of one platform. We refer the readers to Section 3.1 for more discussions.

Benchmarks

We focus on *liquid welfare*, i.e., the total *target-normalized value* to all advertisers under the final allocation, as the measure of efficiency. In the autobidding literature, liquid welfare is used to measure efficiency instead of the usual social welfare because the latter cannot be approximated when advertisers have ROI constraints [2, §2.2]. By the definition of ROI constraints, this is also an upper bound on the total expected revenue that can be generated across platforms.

⁴Due to the stochastic nature of the model (i.e. the value distribution has a positive density almost everywhere in its support), two or more advertisers having the same bid is a zero-measure event.

⁵This is a standard assumption in the autobidding literature (see Aggarwal et al. [2, 5], Alimohammadi et al. [7]). As explained in those papers, due to the high frequency of ad-auctions, autobidders can quickly obtain sharp estimates of other bidders bids, and as such, their bidding problem resembles a complete information game.

DEFINITION 2.3 (LIQUID WELFARE). *Given the allocations $\mathbf{x}$, the liquid welfare is defined as*

$$LW(\mathbf{x}) = \sum_{j \in J} \mathbb{E}_{v_j \sim D_j} \left[\sum_i x_{ij}(v_j) \cdot \frac{v_{ij}}{T_i} \right]$$

The optimal liquid welfare of the system LW^ is thus*

$$LW^* = \sum_{j \in J} \mathbb{E}_{v_j \sim D_j} \left[\max_i \frac{v_{ij}}{T_i} \right]$$

Last, we introduce the notion that captures the level of advertiser competition within a platform, which as we will see, is an important factor that affects the platform's auction choice. It's defined by the ratio of the second-highest value to the highest value. A larger C_j implies higher competition within the platform.

DEFINITION 2.4 (COMPETITION). *For each platform j , define the competition level C_j as $C_j = \frac{E_{v_j \sim D_j} [v_j^{(2)}]}{E_{v_j \sim D_j} [v_j^{(1)}]}$, where $v_j^{(r)}$ is the r -th highest value among $v_{1j}, \dots, v_{nj}$.*

3 The Advertiser Bidding Subgame: Marginal Cost Equalization

In this section, we study the bidding subgame between advertisers under a fixed choice of auction formats for the platforms. All proofs are deferred to Section A.

We first study a single advertiser's bidding problem, as shown in Problem (1). The main result in this section is that the advertisers will set their bid multipliers such that the marginal cost per unit value, henceforth *marginal cost*, is the same across all the platforms they participate in.

As a byproduct of the marginal cost analysis, we define an *elasticity metric*, a normalized version of the marginal cost, that captures the aggressiveness of the pricing of a given auction. It captures how much the advertisers change their bids in the bidding equilibrium when platforms change auctions. This elasticity metric plays a key role in our auction comparator result in the next section.

3.1 Advertiser's Optimal Bidding

We fix a platform j , the auction $A \in \mathcal{A}$, and solve advertiser i 's bidding problem while fixing the bid multipliers of the other advertisers (denoted by $\mu_{-i} = (\mu_{i'j})_{i' \neq i, j \in J}$). We denote by $V_{ij}(\mu_{ij}, \mu_{-i})$ the expected value advertiser i obtains on platform j when using a bid multiplier μ_{ij} . Formally,

$$V_{ij}(\mu_{ij}, \mu_{-i}) = \mathbb{E}_{v_j \sim D_j} [v_{ij} \cdot \mathbb{1}[\mu_{ij}v_{ij} \geq \mu_{kj}v_{kj}, \forall k \neq i]]$$

Similarly, we define the expected cost the advertiser pays as

$$C_{ij}(\mu_{ij}, \mu_{-i}) = \mathbb{E}_{v_j \sim D_j} [p_{ij}(\mu_j \cdot v_j)]$$

We notice that both $V_{ij}(\mu_{ij}, \mu_{-i})$ and $C_{ij}(\mu_{ij}, \mu_{-i})$ are continuous and non-decreasing in μ_{ij} . Throughout this section, we will simplify the notation and use $V_{ij}(\mu_{ij})$ and $C_{ij}(\mu_{ij})$ when μ_{-i} is fixed and clear from context.

Observation 3.1 presented below states that the V_{ij} function has the following shape: As μ_{ij} increases, $V_{ij}(\mu_{ij})$ stays at 0 at the beginning, then is strictly increasing until it hits its maximum, then

becomes flat. The observation follows directly from the fact that the underlying distributions have positive density everywhere in a continuous support. The proof is omitted.

OBSERVATION 3.1. *For every i, j . Let $\underline{\mu}_{ij} = \sup\{\mu_{ij} | V_{ij}(\mu_{ij}) = 0\}$, $\bar{\lambda}_{ij} = \sup\{V_{ij}(\mu_{ij})\}$ and $\bar{\mu}_{ij} = \inf\{\mu_{ij} | V_{ij}(\mu_{ij}) = \bar{\lambda}_{ij}\}$. Then V_{ij} is strictly increasing in $[\underline{\mu}_{ij}, \bar{\mu}_{ij}]$.*

It's clear that the optimal solution in Problem (1) is achieved by $\mu_{ij} \in [\underline{\mu}_{ij}, \bar{\mu}_{ij}]$, $\forall j$. In this range, consider the inverse of the strictly increasing function V_{ij} , denoted by V_{ij}^{-1} . The advertiser's problem can thus be rewritten as the following:

$$\begin{aligned} \max_{\{\lambda_{ij}\}_j} & \sum_j \lambda_{ij} \\ \text{s.t.} & \sum_j C_{ij}(V_{ij}^{-1}(\lambda_{ij})) \leq \sum_j \lambda_{ij} \\ & \lambda_{ij} \in [0, \bar{\lambda}_{ij}], \forall j \end{aligned} \quad (2)$$

We assume without loss of generality that $C_{ij}(V_{ij}^{-1}(\cdot))$ is convex for all i, j so that Problem (2) is a convex problem. In fact, when $C_{ij}(V_{ij}^{-1}(\cdot))$ is not convex, we can replace it by its lower convex envelope, formally

$$\widehat{C_{ij} \circ V_{ij}^{-1}}(\lambda_{ij}) = \min_{\substack{p \in [0,1], \lambda, \lambda' \\ \lambda p + \lambda'(1-p) = \lambda_{ij}}} \{p C_{ij}(V_{ij}^{-1}(\lambda)) + (1-p) C_{ij}(V_{ij}^{-1}(\lambda'))\}$$

We notice that when the advertiser uses a bid multiplier $\mu = V_{ij}^{-1}(\lambda)$ w.p. p and $\mu' = V_{ij}^{-1}(\lambda')$ w.p. $1-p$, she obtains expected value $\lambda p + \lambda'(1-p) = \lambda_{ij}$ and expected cost $p C_{ij}(\mu) + (1-p) C_{ij}(\mu')$. For exposition, we include the convexity assumption of the "value-to-cost" function, in which case the optimal strategy for the advertisers (Problem (1)) is achieved by a deterministic set of bid multipliers.

3.2 Marginal Cost Equalization and Elasticity

To solve the convex problem, we take the Lagrangian dual of Problem (2). The first-order condition shows that $\frac{dC_{ij} \circ V_{ij}^{-1}}{d\lambda_{ij}}$ are equalized at the optimal solution. Mapping back to the μ_{ij} variable, define

$$MC_{ij}(\mu_{ij}, \mu_{-i,j}) = \frac{dC_{ij} \circ V_{ij}^{-1}}{d\lambda_{ij}} = \frac{C'_{ij}(\mu_{ij})}{V'_{ij}(\mu_{ij})}.$$

Theorem 3.2 formally characterizes the optimal solution of Problem (1) (or equivalently Problem (2)). MC_{ij} is the *marginal cost* function, which intuitively corresponds to the additional cost an advertiser would have to pay to marginally increase their value on a platform. The theorem implies that in the advertiser's optimal strategy, the marginal cost are equalized across all platforms where the multiplier is interior, i.e. the advertiser wins partially on this platform. Property (2) of the theorem follows from complementary slackness.

THEOREM 3.2. *Suppose there exists an interior solution of Problem (1) that makes the ROI constraint strictly hold, i.e. $\mu_{ij} \in (\underline{\mu}_{ij}, \bar{\mu}_{ij})$, $\forall j$ such that $\sum_j C_{ij}(\mu_{ij}) < \sum_j V_{ij}(\mu_{ij})$. Then unique optimal solution $\mu_i^* = \{\mu_{ij}^*\}_j$ satisfies:*

- (1) There exists $c > 0$ such that for every platform j , either 1) $MC_{ij}(\mu_{ij}^*, \mu_{-i,j}) = c$; or 2) $MC_{ij}(\mu_{ij}^*, \mu_{-i,j}) > c$ and $\mu_{ij}^* = \underline{\mu}_{ij}$; or 3) $MC_{ij}(\mu_{ij}^*, \mu_{-i,j}) < c$ and $\mu_{ij}^* = \overline{\mu}_{ij}$.
- (2) Either $\sum_j C_{ij}(\mu_{ij}^*) = \sum_j V_{ij}(\mu_{ij}^*)$, i.e. the ROI constraint is tight; or the optimal solution is achieved by $\mu_{ij}^* = \overline{\mu}_{ij}, \forall j$, i.e. advertiser i always wins in all the platforms.

In the rest of the paper, we will focus on advertiser equilibriums where no advertiser wins the entire inventory of one platform, in which case all the bid multipliers are interior and the marginal cost are equalized across all platforms.

Before studying the platform's auction choice and equilibrium, we first define the *elasticity* with respect to an auction format by normalizing $MC_{ij}(\mu_{ij}, \mu_{-i,j})$ by its multiplier μ_{ij} .

DEFINITION 3.3. For a given auction format A , and a set of multipliers μ_{ij} for platform $j \in J$, we define the elasticity of auction A on platform j as follows:

$$\mathcal{E}_{ij}^A(\mu_{ij}, \mu_{-i,j}) = \frac{1}{\mu_{ij}} \cdot MC_{ij}(\mu_{ij}, \mu_{-i,j}) = \frac{C'_{ij}(\mu_{ij})}{\mu_{ij} V'_{ij}(\mu_{ij})}$$

The following lemma shows how the elasticity captures the aggressiveness of the pricing rule of the auction by comparing its value for the SPA and FPA case.

LEMMA 3.4. For any advertiser i , platform j and multiplier profile μ , we have

$$\begin{aligned} \mathcal{E}_{ij}^{SPA}(\mu_{ij}, \mu_{-i,j}) &= 1 \\ \mathcal{E}_{ij}^{FPA}(\mu_{ij}, \mu_{-i,j}) &= 1 + \frac{V_{ij}(\mu)}{\mu_i \cdot \frac{\partial V_{ij}(\mu)}{\partial \mu_i}}. \end{aligned}$$

Observe that the marginal cost on an SPA platform is independent of the bids of other advertisers and only depends on the bid multiplier chosen by the advertiser. On the other hand, on an FPA platform, the marginal cost includes a second positive term which depends on other advertisers' bids. This makes the marginal cost at a given bid multiplier larger for FPA than for SPA.

At the end of this section, we show that the elasticity is scale invariant. This is an important property to obtain our result in the symmetric-advertiser setting.

LEMMA 3.5. For any $c > 0$, $\mathcal{E}_{ij}^A(\mu_{ij}, \mu_{-i,j}) = \mathcal{E}_{ij}^A(c \cdot \mu_{ij}, c \cdot \mu_{-i,j})$.

4 Symmetric Advertisers

In this section, we study the setting where all advertisers are symmetric, or formally $D_{ij} = D_{i',j}$ for all pairs of advertisers i, i' and platform j . We notice that in this I.I.D. setting, by Definition 2.2, advertisers use the same bid multipliers for each platform in any symmetric bidding equilibrium. Thus the allocation is always efficient regardless of the platforms' auction choices, i.e. the advertiser who has the highest value wins the auction.

By Theorem 3.5, the elasticity $\mathcal{E}_{ij}^A(\mu_{ij}, \mu_{-i,j})$ at any symmetric bidding equilibrium equals to the elasticity as if the all advertisers bid their true value (i.e. use a bid multiplier of 1). We thus drop the dependency on the multipliers and index i (due to symmetry) and denote the elasticity at bidding equilibrium as $\mathcal{E}_j(A)$.

We prove the following theorem as the main result of this section. The proofs of Theorem 4.1 is deferred to Section B.1.

THEOREM 4.1. When advertisers are symmetric, for each platform j , and any auction $A_j, A'_j \in \mathcal{A}$, the platform prefers A'_j to A_j regardless of the other platforms' choices if

$$\frac{R_j(A'_j)}{\mathcal{E}_j(A'_j)} \geq \frac{R_j(A_j)}{\mathcal{E}_j(A_j)}$$

Here $R_j(A) = \mathbb{E}_{v_j \sim D_j} [\sum_i p_{ij}(v_j)]$ is the revenue of platform j using auction A , when all advertisers bid their value. Thus, it is a dominant strategy for platform j to choose auction $A_j^* = \arg \max_{A \in \mathcal{A}} \frac{R_j(A)}{\mathcal{E}_j(A)}$. Each platform j choosing auction A_j^* forms a (pure strategy) NE.

4.1 Special Case: SPA vs. FPA

To illustrate the potential sub-optimality of FPA in multi-platform settings, we consider a special setting where $\mathcal{A} = \{SPA, FPA\}$, i.e. the platform can choose between SPA and FPA. We show the platform equilibrium in Theorem 4.2, which is a direct corollary of Theorem 4.1.

THEOREM 4.2. Consider any inefficiency-free setting where the auction set $\mathcal{A} = \{SPA, FPA\}$. For any platform j , if $\mathcal{E}_j(FPA) \cdot C_j > 1$, SPA is the dominant strategy for platform j ; If $\mathcal{E}_j(FPA) \cdot C_j < 1$, FPA is the dominant strategy; If $\mathcal{E}_j(FPA) \cdot C_j = 1$, the platform is indifferent between SPA and FPA.

PROOF. The theorem follows from Theorem 4.1 and the fact that $\mathcal{E}(SPA) = 1$ and $C_j = \frac{R_j(SPA)}{R_j(FPA)}$. $\square$

Theorem 4.2 implies that in this setting, the equilibrium of the game is determined by the following two factors only: the competition level of the game C_j and the elasticity of the advertisers in FPA. An instance with a larger competition level indicates enhanced competition between the advertisers, which makes SPA more favorable. As discussed in Section 3, the elasticity indicates how large the bid change is when a platform changes its auction. Given the competition level of the game, if the elasticity of the advertisers is sufficiently high, then SPA will be the dominant strategy of the game.

We examine the setting with two symmetric advertisers for standard value distributions. Using Theorem 4.2, we show that for any platform, SPA dominates FPA if the advertisers' value distribution for this platform is either a uniform distribution $U[a, b](a > 0)$ or an exponential distribution. See Section B.2 for details.

5 Asymmetric Advertisers

In this section, we consider the general case where advertisers have asymmetric value distributions, and thus the allocation may become inefficient at equilibrium for some choices of auction formats. Now a platform faces a trade-off across efficiency loss, competition and elasticity when choosing among different auction formats. While solving the problem in its full generality is complex and no simple analytical solution seems to exist, we focus on the special setting considered in Section 4.1 and provide sufficient conditions for SPA to dominate FPA and vice versa. Due to page restrictions, all proofs in this section are deferred to Section C.1.

We consider the setting where two asymmetric advertisers compete in two symmetric platforms, and each platform can choose between FPA and SPA. Intuitively, since FPA with uniform bidding

allocates queries more efficiently than SPA, FPA should be relatively more attractive to platforms in the asymmetric-advertiser case than in the symmetric-advertiser case (FPA is likely to lose less efficiency than SPA). As such, the conditions of Theorem 4.2 (appropriately modified for the asymmetric-advertiser case) should suffice for (FPA, FPA) to be a Nash equilibrium of the game. This is indeed the case. In the following theorem, we give sufficient conditions for (FPA, FPA) to be an equilibrium by generalizing the conditions in Theorem 4.2 to the asymmetric-advertiser case.

THEOREM 5.1. *(FPA, FPA) is a NE of the game if for both advertisers $i = 1, 2$, it holds that $\mathcal{E}_i^{FPA}(\mu^*) \cdot \frac{\mathbb{E}[v_i \cdot \mathbb{1}[v_i \leq v_{-i}]]}{\mathbb{E}[v_i \cdot \mathbb{1}[v_i > v_{-i}]]} < 1$. Here μ^* is the advertisers' bid multipliers in the bidding equilibrium of the (SPA, FPA) profile. v_{-i} is the value for the other advertiser. We note that when both advertiser are symmetric, the above condition reduces to the condition in Theorem 4.2.*

Next, we study conditions under which (SPA, SPA) is an equilibrium. Here, the loss of efficiency makes SPA less attractive to the platform (as SPA tends to lose more efficiency than FPA). Hence, the conditions in Theorem 4.2 are not sufficient for SPA to be chosen at equilibrium. We define two quantities to help us pin down sufficient conditions: H indicates how differently the two advertisers bid in the FPA platform in the (SPA, FPA) subgame. A larger H implies a larger difference between the bid multipliers at FPA platform, which implies a less efficient allocation; δ indicates the efficiency loss of the (SPA, SPA) subgame. Note that $H = 1$ and $\delta = 0$ when advertisers are symmetric. The next theorem gives sufficient conditions for (SPA, SPA) to be an equilibrium. We see that as the efficiency loss, as captured by H and δ , increases, it becomes harder to satisfy the conditions for SPA to be chosen at equilibrium.

THEOREM 5.2. *Let $\mu^* = \{\mu_{ij}^*\}_{i,j \in \{1,2\}}$ be the advertisers' bid multipliers in the bidding equilibrium of the (SPA, FPA) profile. Let $H = \frac{\max\{\mu_{12}^*, \mu_{22}^*\}}{\min\{\mu_{12}^*, \mu_{22}^*\}} \geq 1$. Let $\delta \in [0, 1)$ be the fraction of the efficiency that is lost in the (SPA, SPA) profile, i.e. $REV_1(SS) + REV_2(SS) = (1 - \delta) \cdot (REV_1(FF) + REV_2(FF))$.⁶ Then (SPA, SPA) is a NE of the game if*

$$\min\{\mathcal{E}_1^{FPA}(\mu^*), \mathcal{E}_2^{FPA}(\mu^*)\} \cdot C \geq H \cdot \frac{1 + \delta}{1 - \delta}$$

We note that when both advertiser are symmetric, the above condition reduces to the condition in Theorem 4.2.

6 A Case Study: Uniform vs Constant Advertisers

We consider two symmetric platforms and two advertisers, one with valuation $v_1 \sim U[0, \alpha]$ (for some $\alpha \in (2, 4)$) and v_2 drawn from a point mass distribution with value 1. We restrict our attention to the case where $\alpha \in (2, 4)$ because for values outside this range, the equilibria of the game result in non-interior solutions.

CLAIM 6.0.1. *For $\alpha \notin (2, 4)$, in any advertisers' equilibrium of the (SPA, SPA) auction profile, one advertiser will win all queries in both platforms.*

⁶We notice that the total efficiency equals to the total revenue of the platforms since the ROS constraint is met for advertisers.

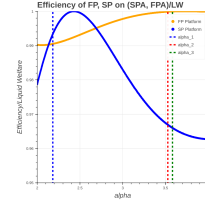


Figure 1: This plot shows the efficiency of the FP and SP platforms on the (SPA, FPA) subgame.

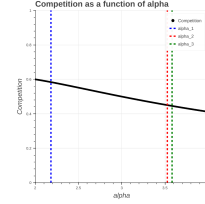


Figure 2: This plot shows the competition metric C as a function α for the case of a linear and constant-valued advertiser.

For tractability purposes, we restrict the platforms auction choice set $\mathcal{A}$ to FPA and SPA. We now state the main result of this section. Due to space constraints, we will defer the proof of Theorem 6.1 to the full version of the paper.

THEOREM 6.1. *The platforms' equilibria of the game in this setting depend on the value of α as follows. There exist $\alpha_1 \leq \alpha_2 \leq \alpha_3 \leq \alpha_4$ s.t.:*

- for $\alpha \in (2, \alpha_1)$ there are three Nash equilibria: (SPA, FPA), (FPA, SPA) and a mixture of both,
- for $\alpha \in [\alpha_1, \alpha_2)$, there is a unique Nash equilibrium: (SPA, SPA),
- for $\alpha \in [\alpha_2, \alpha_3)$, there are three Nash equilibria: (FPA, FPA), (SPA, SPA) and a mixture of both,
- for $\alpha \in [\alpha_3, 4)$, there is a unique Nash equilibrium: (FPA, FPA).

Observe the plots of efficiency, competition and elasticities as a function of the parameter α (see Figures 1, 2, 3). As α increases, there is a sharp loss of efficiency on the SPA platform relative to the FPA platform. This, combined with a decrease in advertiser competition C , is sufficient to overcome an increase in elasticity and shift the equilibrium from (SPA, SPA) to (FPA, FPA) (for high values of α). This observation contrasts with the results from Section 4 where advertiser competition and elasticity were the only driving factors in determining equilibria. Here we observe that efficiency losses may also play a substantial role in determining equilibria, as shown also in Theorem 5.2.

6.1 Impact of Market Share on Auction Choice

In this subsection, we present some observations based on the parametric family of examples described above, with platforms with identical inventories but asymmetric market share. We introduce an additional parameter $\gamma \in [0, 1]$ which specifies the market share of the first platform. We solve for the equilibria in this setting as a

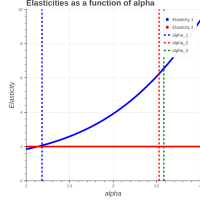


Figure 3: This plot shows the elasticity of both advertisers as a function α for the case of a linear and constant-valued advertiser.

function of both the first platform's market share (γ) and the first advertiser's valuation function (parameterized by α). See Figure 4.

The analysis shows that as a platform's market share increases, the platform is more likely to use FPA compared to SPA. This matches with the intuition that a larger platform is less worried about substitution effect and more worried with allocating efficiently in order to generate more revenue. Conversely, we also observe that, for many values of the parameter α , when each platform has a substantial market share (say $\gamma \in [.25, .75]$), the first platform will choose a Second Price format. This matches with the intuition that in competitive environments, platforms are willing to give up on efficient allocations in order to prevent being substituted by their competition. We also observe that once the platform with the larger market share prefers to use FPA, it will continue to prefer FPA as its market share increases. Conversely, once the platform with the smaller market share prefers to use SPA, it will continue to use SPA as its market share decreases.

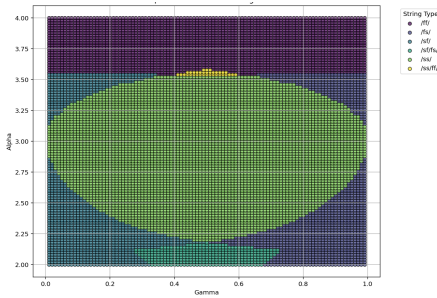


Figure 4: Plot describing the platform equilibrium in the parametric example described in Appendix 6.

7 Importance of Advertisers' Elasticity

Results in previous sections show that the platform equilibrium depends heavily on advertiser elasticity (i.e., responsiveness to auction changes). In this section, we present two results that reinforce the importance of the advertisers' elasticity in allowing for equilibria other than the one where all platforms use FPA. When a platform changes its auction, the advertisers will solve Problem (1) and may change their bid multipliers in reaction to the auction change. For each of the results shown in this section, we disable the advertisers' reaction to an auction change to some degree, and find that the platforms always end up in the all-FPA equilibrium.

In the first result, advertisers are *forced* to use the same bid multiplier across platforms, and thus cannot equalize marginal costs;⁷ In the second result, there is a single *strategic* advertiser bidding across platforms and competitors are *static* advertisers that always bid their value and do not adjust their bid multipliers when the auction changes.

7.1 Sub-optimal Uniform Bidding Across Platforms

We first study the case where advertisers are forced to use the same bid multiplier across all platforms, i.e. advertiser i 's bid in platform j is given by $\mu_i v_{ij}$ for all $j \in J$. We show that, if advertisers are using *weakly undominated*⁸ bidding strategies, then in the unique Nash equilibrium of the game, each platform uses FPA and, in return, each advertiser uses a bid multiplier of 1 across all platforms. Removing weakly dominated strategies, those where there are weakly dominant alternatives, precludes unnatural equilibria. For example, an advertiser who is bidding below their value and getting no value would increase their multiplier until they either bid their value their expected value increases (Claim D.0.1). Furthermore, the revenue across all platforms matches the liquid welfare W^* . In particular, if a single entity owns all platforms, then using FPA on all platforms maximizes the revenue. The proof of Theorem 7.1 is deferred to Appendix D.

THEOREM 7.1. *The unique equilibrium of the platform-advertiser game under uniform bidding is for the platforms to choose FPA and for advertisers to submit multipliers $\mu_i = 1$. Furthermore, the total revenue across all platforms is equal to the liquid welfare W^* .*

7.2 Single Strategic Advertiser vs. Static Advertisers

In this section, we study the setting where a single strategic advertiser bids across multiple platforms. Motivated by practical considerations, we assume the platforms can only choose between FPA and SPA. The strategic advertiser competes with static advertisers that always bid their value, i.e. use a bid multiplier of 1. We show that for this case, FPA weakly dominates SPA for all platforms.

THEOREM 7.2. *In the setting with single strategic advertiser and static advertisers, FPA weakly dominates SPA for all platform j .*

PROOF. Fix any platform j . Since static advertisers will always bid their value, the single strategic advertiser is competing with some fixed random variable v_j for each platform j (which is the largest value among all static advertisers). We notice that if a platform j uses FPA, the expected revenue must be at least $\mathbb{E}(v_j)$: If the strategic advertiser wins on the platform, their bid (and payment) must be at least v_j ; If the static advertiser wins, the payment is v_j . On the other hand, if the platform uses SPA, then the payment at each bid profile is at most v_j , which makes the expected revenue at most $\mathbb{E}(v_j)$. Thus FPA is a weakly dominant strategy. $\square$

⁷If all the platforms are using SPA, note that using the same bid multiplier is optimal [3], and hence, it equalizes the marginal cost by Theorem 3.2.

⁸Weakly undominated strategies are those that are not weakly dominated by any other strategy.

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Appendix

A Missing Details from Section 3

PROOF OF THEOREM 3.2. The min-version of problem in Equation (2):

$$\begin{aligned} \min_{\{\lambda_{ij}\}_j} & - \sum_j \lambda_{ij} \\ \text{s.t.} & \sum_j C_{ij}(V_{ij}^{-1}(\lambda_{ij})) \leq \sum_j \lambda_{ij} \\ & \lambda_{ij} \in [0, \bar{\lambda}_{ij}], \forall j \end{aligned} \quad (3)$$

Let $g_{ij} = C_{ij} \circ V_{ij}^{-1}$. Consider the Lagrangian dual of the problem by Lagrangifying the ROI constraint. Let α be the dual variable. Then

$$\begin{aligned} L(\lambda_i, \alpha) &= - \sum_j \lambda_{ij} + \alpha \cdot \left(\sum_j g_{ij}(\lambda_{ij}) - \sum_j \lambda_{ij} \right) \\ &= \sum_j [\alpha g_{ij}(\lambda_{ij}) - (1 + \alpha)\lambda_{ij}] \end{aligned}$$

Firstly, strong duality holds by the Slater's condition: there exists an interior solution $\lambda_{ij} \in (0, \bar{\lambda}_{ij})$, $\forall j$ where the ROI constraint is not tight.

$$q(\alpha) = \inf_{\lambda_{ij} \in [0, \bar{\lambda}_{ij}], \forall j} L(\lambda_i, \alpha) = \sum_j \min_{\lambda_{ij} \in [0, \bar{\lambda}_{ij}]} [\alpha g_{ij}(\lambda_{ij}) - (1 + \alpha)\lambda_{ij}]$$

Let λ_i and α^* be the optimal primal and dual solution respectively. If $\alpha^* = 0$, then $L(\lambda_i, \alpha^*)$ is minimized when $\lambda_{ij}^* = \bar{\lambda}_{ij}$, $\forall j$.

When $\alpha^* > 0$, then to minimize $L(\lambda_i, \alpha^*)$, we have that $\lambda_{ij}^* = \arg \min_{\lambda_{ij} \in [0, \bar{\lambda}_{ij}]} [\alpha^* g_{ij}(\lambda_{ij}) - (1 + \alpha^*)\lambda_{ij}]$

Taking the derivative of $\alpha^* g_{ij}(\lambda_{ij}) - (1 + \alpha^*)\lambda_{ij}$, we have that for interior points $\lambda_{ij}^* \in (0, \bar{\lambda}_{ij})$, $g'_{ij}(\lambda_{ij}^*) = (1 + \alpha^*)/\alpha^*$. Otherwise, when $g'_{ij}(0) > (1 + \alpha^*)/\alpha^*$, $\lambda_{ij}^* = 0$; when $g'_{ij}(\bar{\lambda}_{ij}) < (1 + \alpha^*)/\alpha^*$, $\lambda_{ij}^* = \bar{\lambda}_{ij}$. Moreover, by complementary slackness we have that $\sum_j g_{ij}(\lambda_{ij}^*) = \sum_j \lambda_{ij}^*$.

Mapping the above statements back to the multiplier space via $\mu_{ij} = V_{ij}^{-1}(\lambda_{ij})$ completes the proof. $\square$

PROOF OF LEMMA 3.5. Due to Definition 2.1, we know that for any auction A we choose, the payment is homogeneous on the set of multipliers, i.e., $p(c \cdot \mathbf{b}) = c \cdot p(\mathbf{b})$. By linearity of expectations $C_{ij}(c \cdot \mu_j) = c \cdot C_{ij}(\mu_j)$. On the other hand, the expected value remains the same if we scale all bids down by c . Since the auction always awards the queries to the highest bidder, the value of the allocation for every advertiser will remain the same if all bids are scaled down by a some constant c . Therefore, $V_{ij}(c \cdot \mu_j) = V_{ij}(\mu_j)$. Plugging all of this into the definition of elasticity, we get the desired result. $\square$

PROOF OF LEMMA 3.4. When the auction is SPA, from the Myerson characterization of truthful auctions, we have that $C_{ij}(\mu_{ij}) = \mu_{ij}V(\mu_{ij}) - \int_0^{\mu_{ij}} V(z)dz$. Then, $\mathcal{E}_{ij}^{SPA}(\mu_{ij}, \mu_{-i,j}) = \frac{C_{ij}(\mu_{ij})}{\mu_{ij}V'_{ij}(\mu_{ij})} = 1$ immediately follows since $C'_{ij}(\mu_{ij}) = \mu_{ij}V'_{ij}(\mu_{ij})$.

If the auction is FPA, then $C_{ij}(\mu_{ij}) = \mu_{ij}V(\mu_{ij})$ and therefore $C'_{ij}(\mu_{ij}) = \mu_{ij}V'_j(\mu_{ij}) + V_{ij}(\mu_{ij})$. Dividing both sides by $\mu_{ij}V'_{ij}(\mu_{ij})$ yields

$$\mathcal{E}_{ij}^{FPA}(\mu_{ij}) = 1 + \frac{V_{ij}(\mu_{ij})}{\mu_{ij}V'_{ij}(\mu_{ij})},$$

which completes the proof of the Lemma. $\square$

B Missing Details from Section 4

B.1 Missing Proofs from Section 4

PROOF OF THEOREM 4.1. Fix any platform j and any set of auction choices for all the other platforms, denoted by $A_{-j} = \{A_k\}_{k \neq j}$. Consider an auction choice where platform j chooses auction A . Denote by $\mu_k(A, A_{-j})$ the multiplier of all advertisers on platform k in the (symmetric) bidding equilibrium. Denote by $\text{REV}_k(A, A_{-j})$ the revenue of any platform $k \in [m]$. We will prove that $\text{REV}_j(A_j, A_{-j}) \geq \text{REV}_j(A_j, A_{-j}), \forall A_j, A'_j \in \mathcal{A}$ such that $\frac{R_j(A'_j)}{\mathcal{E}_j(A'_j)} \geq \frac{R_j(A_j)}{\mathcal{E}_j(A_j)}$.

Note that when advertisers are symmetric, the total welfare stays the same for any platforms' auction choices (which is the optimal welfare). Thus total revenue across all platforms (i.e. the total payment from advertisers) stays the same since the ROI constraints for all advertisers are tight. In other words,

$$\begin{aligned} & \text{REV}_j(A'_j, A_{-j}) + \sum_{k \neq j} \text{REV}_k(A'_j, A_{-j}) \\ &= \text{REV}_j(A_j, A_{-j}) + \sum_{k \neq j} \text{REV}_k(A_j, A_{-j}) \end{aligned}$$

Thus, we have

$$\begin{aligned} & \text{REV}_j(A'_j, A_{-j}) \geq \text{REV}_j(A_j, A_{-j}) \\ \iff & \frac{\text{REV}_j(A'_j, A_{-j})}{\sum_{k \neq j} \text{REV}_k(A'_j, A_{-j})} \geq \frac{\text{REV}_j(A_j, A_{-j})}{\sum_{k \neq j} \text{REV}_k(A_j, A_{-j})} \\ \iff & \frac{\mu_j(A'_j, A_{-j}) \cdot R_j(A'_j)}{\sum_{k \neq j} \mu_k(A'_j, A_{-j}) \cdot R_k(A_k)} \geq \frac{\mu_j(A_j, A_{-j}) \cdot R_j(A_j)}{\sum_{k \neq j} \mu_k(A_j, A_{-j}) \cdot R_k(A_k)} \end{aligned}$$

where the last step comes from the homogeneity of the auction. By Theorem 3.2, $\frac{\mu_k(A, A_{-j})}{\mu_j(A, A_{-j})} = \frac{\mathcal{E}_j(A)}{\mathcal{E}_k(A_k)}$ for any auction $A \in \mathcal{A}$. Thus

$$\begin{aligned} & \frac{\mu_j(A'_j, A_{-j}) \cdot R_j(A'_j)}{\sum_{k \neq j} \mu_k(A'_j, A_{-j}) \cdot R_k(A_k)} \geq \frac{\mu_j(A_j, A_{-j}) \cdot R_j(A_j)}{\sum_{k \neq j} \mu_k(A_j, A_{-j}) \cdot R_k(A_k)} \\ \iff & \frac{R_j(A'_j)}{\mathcal{E}_j(A'_j) \cdot \sum_{k \neq j} \frac{R_k(A_k)}{\mathcal{E}_k(A_k)}} \geq \frac{R_j(A_j)}{\mathcal{E}_j(A_j) \cdot \sum_{k \neq j} \frac{R_k(A_k)}{\mathcal{E}_k(A_k)}} \\ \iff & \frac{R_j(A'_j)}{\mathcal{E}_j(A'_j)} \geq \frac{R_j(A_j)}{\mathcal{E}_j(A_j)} \end{aligned}$$

This completes the proof. $\square$

B.2 Equilibrium for Classic Valuations in the Symmetric-Advertiser Setting

We apply the result of Theorem 4.2 to show that SPA strongly dominates FPA for classic valuations.

Fix any platform j . When there are two advertisers, the elasticity for FPA can be simplified as follows (we omit index j as the platform is fixed).

$$\begin{aligned} V_1(\mu_1, \mu_2) &= \int_0^\infty v_1 f_1(v_1) F_2\left(\frac{\mu_1 v_1}{\mu_2}\right) dv_1 \\ \frac{\partial V_1(\mu_1, \mu_2)}{\partial \mu_1} &= \frac{1}{\mu_2} \cdot \int_0^\infty v_1^2 f_1(v_1) f_2\left(\frac{\mu_1 v_1}{\mu_2}\right) dv_1 \\ \mathcal{E}_1^{FPA}(\mu_1, \mu_2) &= 1 + \frac{\mu_2}{\mu_1} \cdot \frac{\int_0^\infty v_1 f_1(v_1) F_2\left(\frac{\mu_1 v_1}{\mu_2}\right) dv_1}{\int_0^\infty v_1^2 f_1(v_1) f_2\left(\frac{\mu_1 v_1}{\mu_2}\right) dv_1} \end{aligned}$$

With the advertiser symmetry, it can be further simplified as:

$$\begin{aligned} \mathcal{E}(FPA) &= \mathcal{E}^{FPA}(1, 1) = 1 + \frac{\int_0^\infty v f(v) F(v) dv}{\int_0^\infty v^2 f^2(v) dv} \\ C &= \frac{E[\min\{v_1, v_2\}]}{E[\max\{v_1, v_2\}]} = \frac{\int_0^\infty v f(v) (1 - F(v)) dv}{\int_0^\infty v(f(v) F(v)) dv} \end{aligned}$$

We will verify as follows that $\mathcal{E}(FPA) \cdot C > 1$ for uniform and exponential distributions. This implies that SPA dominates FPA when the value distributions for all platforms are either uniform or exponential. Note that we do not require those value distributions to be identical.

B.2.1 Uniform Distribution. For a uniform distribution $U[a, b]$ ($a > 0$), $f(v) = \frac{1}{b-a}$, $F(v) = \frac{v-a}{b-a}$. With some basic algebra, we have

$$\begin{aligned} \mathcal{E}(FPA) &= 1 + \frac{\int_a^b v \cdot \frac{1}{b-a} \cdot \frac{v-a}{b-a} dv}{\int_a^b \frac{v^2}{(b-a)^2} dv} = 1 + \frac{\frac{2b+a}{6}}{\frac{b^2+ab+a^2}{3(b-a)}} = \frac{4b^2 + ab + a^2}{2(b^2 + ab + a^2)} \\ C &= \frac{\frac{2a+b}{3}}{\frac{a+2b}{3}} = \frac{2a+b}{a+2b} \end{aligned}$$

$$\mathcal{E}(FPA) \cdot C - 1 = \frac{3ab(b-a)}{2(b^2 + ab + a^2)(a+2b)} > 0$$

B.2.2 Exponential Distribution. Consider exponential distribution with parameter $\lambda > 0$. $f(v) = \lambda e^{-\lambda v}$, $F(v) = 1 - e^{-\lambda v}$.

$$\int_0^\infty v f(v) F(v) dv = \lambda \int_0^\infty (v e^{-\lambda v} - v e^{-2\lambda v}) dv = \frac{3}{4\lambda}$$

$$\begin{aligned} \int_0^\infty v^2 f^2(v) dv &= \lambda^2 \int_0^\infty v^2 e^{-2\lambda v} dv \\ &= \lambda^2 \cdot \frac{1}{(2\lambda)^3} \Gamma(2) = \frac{1}{4\lambda} \end{aligned}$$

where $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt$ is the Gamma function. Thus

$$\mathcal{E}(FPA) = 1 + \frac{\int_0^\infty v f(v) F(v) dv}{\int_0^\infty v^2 f^2(v) dv} = 1 + \frac{3/(4\lambda)}{1/(4\lambda)} = 4$$

For C , notice that $\min\{v_1, v_2\}$ follows from exponential distribution with parameter 2λ , we have

$$\mathbb{E}[\min\{v_1, v_2\}] = \frac{1}{2\lambda}$$

Also we have

$$\mathbb{E}[\max\{v_1, v_2\}] = \mathbb{E}[v_1] + \mathbb{E}[v_2] - \mathbb{E}[\min\{v_1, v_2\}] = \frac{3}{2\lambda}$$

$$\text{Thus } C = \frac{1/(2\lambda)}{3/(2\lambda)} = \frac{1}{3}.$$

$$\mathcal{E}(FPA) \cdot C = \frac{4}{3} > 1$$

C Missing Details from Section 5

C.1 Missing Proofs from Section 5

PROOF OF LEMMA 5.1. For any auction A and B, we denote by $\text{REV}_1(AB)$, $\text{REV}_2(AB)$ the revenue of platform 1 and 2 respectively, when platform 1 chooses A and platform 2 chooses B. In the notation we will use “S” to stand for SPA and “F” to stand for FPA.

We will show that $\text{REV}_1(FF) \geq \text{REV}_1(SF)$ and $\text{REV}_2(FF) \geq \text{REV}_2(FS)$, which implies that (FPA, FPA) is a NE.

We first prove $\text{REV}_1(FF) \geq \text{REV}_1(SF)$ by contradiction. Suppose $\text{REV}_1(FF) < \text{REV}_1(SF)$. We notice that since platforms are symmetric, when both platforms choose FPA, both advertisers bid their true value, which results in an efficient allocation. Thus $\text{REV}_1(FF) + \text{REV}_2(FF) \geq \text{REV}_1(SF) + \text{REV}_2(SF)$.

Hence, we have $\text{REV}_2(FF) > \text{REV}_2(SF)$ and $\frac{\text{REV}_1(SF)}{\text{REV}_2(SF)} > \frac{\text{REV}_1(FF)}{\text{REV}_2(FF)} = 1$, where the last equality comes from the fact that platforms are symmetric.

Denote by μ_{11}, μ_{12} the multiplier of advertiser 1 for both platforms in the (SPA, FPA) profile. And μ_{11}, μ_{12} the multiplier of advertiser 2. Then

$$\begin{aligned} \text{REV}_1(SF) &= \mathbb{E}[\min\{\mu_{11}v_1, \mu_{21}v_2\}] \\ &\leq \mu_{11} \cdot \mathbb{E}[v_1 \mathbb{I}[v_1 \leq v_2]] + \mu_{21} \cdot \mathbb{E}[v_2 \mathbb{I}[v_2 \leq v_1]] \end{aligned}$$

$$\begin{aligned} \text{REV}_2(SF) &= \mathbb{E}[\max\{\mu_{12}v_1, \mu_{22}v_2\}] \\ &\geq \mu_{12} \cdot \mathbb{E}[v_1 \mathbb{I}[v_1 > v_2]] + \mu_{22} \cdot \mathbb{E}[v_2 \mathbb{I}[v_2 > v_1]] \end{aligned}$$

We notice that by the definition of elasticity and Theorem 3.2, $\frac{\mu_{11}}{\mu_{12}} = \frac{\mathcal{E}_1^{FPA}(\mu^*)}{\mathcal{E}_2^{FPA}(\mu^*)} = \mathcal{E}_1^{FPA}(\mu^*)$, where the last inequality is by Theorem 3.4. Similarly, $\mathcal{E}_2^{FPA}(\mu^*) = \frac{\mu_{21}}{\mu_{22}}$. Thus

$$\begin{aligned} \text{REV}_1(SF) &\leq \mu_{11} \cdot \mathbb{E}[v_1 \mathbb{I}[v_1 \leq v_2]] + \mu_{21} \cdot \mathbb{E}[v_2 \mathbb{I}[v_2 \leq v_1]] \\ &= \mathcal{E}_1^{FPA}(\mu^*) \mu_{12} \cdot \mathbb{E}[v_1 \mathbb{I}[v_1 \leq v_2]] + \mathcal{E}_2^{FPA}(\mu^*) \mu_{22} \cdot \mathbb{E}[v_2 \mathbb{I}[v_2 \leq v_1]] \\ &\leq \mu_{12} \cdot \mathbb{E}[v_1 \mathbb{I}[v_1 > v_2]] + \mu_{22} \cdot \mathbb{E}[v_2 \mathbb{I}[v_2 > v_1]] \leq \text{REV}_2(SF) \end{aligned}$$

where the last inequality comes from the conditions in the theorem statement. This contradicts with $\frac{\text{REV}_1(SF)}{\text{REV}_2(SF)} > 1$. Thus we have proved $\text{REV}_1(FF) \geq \text{REV}_1(SF)$.

The other inequality $\text{REV}_2(FF) \geq \text{REV}_2(FS)$ can be proved analogously by switching the role of both platforms.

When advertisers are symmetric, we notice that $\mathcal{E}_1^{FPA}(\mu^*) = \mathcal{E}_2^{FPA}(\mu^*) = \mathcal{E}^{FPA}(1)$, and

$$\begin{aligned} \frac{E[v_i \mathbb{I}[v_i \leq v_{-i}]]}{E[v_i \mathbb{I}[v_i > v_{-i}]]} &= \frac{1/2(E[v_1 \mathbb{I}[v_1 \leq v_2]] + E[v_1 \mathbb{I}[v_1 \leq v_2]])}{1/2(E[v_1 \mathbb{I}[v_1 > v_2]] + E[v_1 \mathbb{I}[v_1 > v_2]])} \\ &= \frac{\mathbb{E}[\min\{v_1, v_2\}]}{\mathbb{E}[\max\{v_1, v_2\}]} \end{aligned}$$

Thus the condition is reduced to the condition in Theorem 4.2. $\square$

PROOF OF THEOREM 5.2. We show that $\text{REV}_1(SS) \geq \text{REV}_1(FS)$ and $\text{REV}_2(SS) \geq \text{REV}_2(SF)$, which implies that (SPA, SPA) is a NE. We will prove $\text{REV}_2(SS) \geq \text{REV}_2(SF)$ by contradiction. The other inequality is analogous due to platform symmetry. Suppose $\text{REV}_2(SS) < \text{REV}_2(SF)$. We have

$$\begin{aligned} \text{REV}_1(SS) + \text{REV}_2(SS) &= (1 - \delta) \cdot (\text{REV}_1(FF) + \text{REV}_2(FF)) \\ &\geq (1 - \delta) \cdot (\text{REV}_1(SF) + \text{REV}_2(SF)) \end{aligned}$$

Also $\text{REV}_1(SS) = \text{REV}_2(SS)$ as platforms are symmetric. Thus

$$\text{REV}_1(SF) < \frac{1+\delta}{1-\delta} \cdot \text{REV}_1(SS), \quad \frac{\text{REV}_2(SF)}{\text{REV}_1(SF)} > \frac{1+\delta}{1-\delta}.$$

By definition of the revenue of the platforms, we have

$$\begin{aligned} \frac{\text{REV}_2(SF)}{\text{REV}_1(SF)} &= \frac{\mathbb{E}[\max\{\mu_{12}^* \cdot v_1, \mu_{22}^* \cdot v_2\}]}{\mathbb{E}[\min\{\mu_{11}^* \cdot v_1, \mu_{21}^* \cdot v_2\}]} \\ &\leq \frac{\max\{\mu_{12}^*, \mu_{22}^*\} \cdot \mathbb{E}[\max\{v_1, v_2\}]}{\min\{\mu_{11}^*, \mu_{21}^*\} \cdot \mathbb{E}[\min\{v_1, v_2\}]} \end{aligned}$$

Recall that $\mathcal{E}_1^{FPA}(\mu^*) = \frac{\mu_{11}^*}{\mu_{12}^*}$, $\mathcal{E}_2^{FPA}(\mu^*) = \frac{\mu_{21}^*}{\mu_{22}^*}$ by definition of elasticity.

Thus

$$\begin{aligned} \frac{\max\{\mu_{12}^*, \mu_{22}^*\}}{\min\{\mu_{11}^*, \mu_{21}^*\}} &= \frac{\max\{\mu_{12}^*, \mu_{22}^*\}}{\min\{\mathcal{E}_1^{FPA}(\mu^*) \mu_{12}^*, \mathcal{E}_2^{FPA}(\mu^*) \mu_{22}^*\}} \\ &\leq \frac{1}{\min\{\mathcal{E}_1^{FPA}(\mu^*), \mathcal{E}_2^{FPA}(\mu^*)\}} \cdot \frac{\max\{\mu_{12}^*, \mu_{22}^*\}}{\min\{\mu_{12}^*, \mu_{22}^*\}} \\ &= \frac{H}{\min\{\mathcal{E}_1^{FPA}(\mu^*), \mathcal{E}_2^{FPA}(\mu^*)\}} \end{aligned}$$

We notice that $C = \frac{\mathbb{E}[\min\{v_1, v_2\}]}{\mathbb{E}[\max\{v_1, v_2\}]}$. Thus

$$\frac{1 - \delta}{1 + \delta} < \frac{\text{REV}_2(SF)}{\text{REV}_1(SF)} \leq \frac{H}{\min\{\mathcal{E}_1^{FPA}(\mu^*), \mathcal{E}_2^{FPA}(\mu^*)\} \cdot C}$$

which contradicts with the assumption in the theorem statement. Thus $\text{REV}_2(SS) \geq \text{REV}_2(SF)$.

Finally, when advertisers are symmetric, we have that $H = 1$ and $\delta = 0$. One can easily verify that the condition in the statement reduces to the one in Theorem 4.2. $\square$

D Missing Proofs From Section 7

PROOF OF THEOREM 7.1. We first prove the following claim.

CLAIM D.0.1. *For any choice of auctions by the platforms, any strategy where advertiser i uses a multiplier $\mu_i < 1$ is weakly dominated by a strategy where they use a multiplier of $\mu_i = 1$.*

PROOF. Suppose advertiser i bids using $\mu_i < 1$. Since the auctions we consider will not charge the winner more than their bid, the total payment is at most $\sum_j \mu_i v_{ij}$ which is strictly less than their value $\sum_j v_{ij}$. By using $\mu_i = 1$ the advertiser could have weakly more

expected value than with μ_i without violating their ROI constraint. $\square$

We will prove the theorem when advertisers are using undominated strategies. Let

$$LW_j^* = \mathbb{E}_{v_j \sim D_j} \left[\max_i v_{ij} \right],$$

which is the liquid welfare that platform j contributes to the liquid welfare W^* . Recall that $LW^* = \sum_{j \in J} LW_j^*$.

We claim that in any equilibrium, the revenue platform j gets, Rev_j , satisfies that $\text{Rev}_j \geq LW_j^*$. Indeed because advertisers are using undominated bidding strategies, if platform j uses an FPA then the revenue it would get at each value profile, which is $\max_{i \in I} \mu_i v_{ij}$, is no less than $\max_{i \in I} v_{ij}$ since $\mu_i \geq 1$ for all $i \in I$. Thus, $\text{Rev}_j \geq LW_j^*$, since at equilibrium, the platform's revenue must be at least the one when it uses FPA. Also by the ROI constraint, $\sum_j \text{Rev}_j \leq LW^* = \sum_j LW_j^*$. Thus $\text{Rev}_j = LW_j^*, \forall j$. Hence, FPA is the unique equilibrium of the game. $\square$